

Analytic Geometry

Slope · Graphing Lines · Parallel & Perpendicular Lines · Length of a Segment · Midpoint

MTH1W · Class — Guided Practice

Name: _____

Date: _____

Goal: Find slope, graph and compare lines, and compute segment length and midpoint on the Cartesian plane.

Warm-up (quick recall)

1. State the slope of the line $y = 4x - 1$.
2. State the y-intercept of the line $y = 2x + 9$.
3. Find the slope of the line through (0, 1) and (3, 7).

Examples & Practice

For each skill, study the worked **Example**, then solve the questions that follow. Difficulty rises within each set (L1 → L3). Answers are on the last page.

Slope of a Line

Example:

Find the slope of the line through A(2, 3) and B(6, 11).

$$m = (y_2 - y_1) / (x_2 - x_1) = (11 - 3) / (6 - 2) = 8 / 4.$$

= 2. Answer: slope = 2.

4. Find the slope of the line $y = 5x - 2$. **[L1]**

5. Find the slope of the line through P(1, 2) and Q(5, 10). **[L1]**
6. Find the slope of the line through (-3, 4) and (3, -8). **[L2]**
7. The line through (k, 3) and (6, 11) has slope 4. Find the value of k. **[L3]**

Graphing Lines (describing the graph)

Example:

Describe the graph of $y = 2x - 4$.

Slope = 2 (up 2, right 1); y-intercept (0, -4); x-intercept: $0 = 2x - 4 \rightarrow (2, 0)$.

Two points on the line: (0, -4) and (2, 0).

8. For $y = x + 3$, state the slope and the y-intercept. **[L1]**

9. For $y = -2x + 6$, state the y-intercept and find the x-intercept. **[L1]**
10. For $y = 3x - 6$, state the slope, the y-intercept, and two points on the line. **[L2]**
11. Describe the graph of $2x + y = 4$: give slope, y-intercept and two plotted points. **[L3]**

Parallel Lines (equal slopes)

Example:

Are $y = 3x + 1$ and $y = 3x - 5$ parallel?
Both lines have slope 3; the slopes are equal.
Yes — parallel (same slope, different y-intercept).

12. State the slope of any line parallel to $y = 4x + 2$. **[L1]**

13. Are $y = 2x + 7$ and $y = 2x - 1$ parallel? Explain. **[L1]**
14. Write the equation of the line parallel to $y = -3x + 5$ that passes through $(0, 2)$. **[L2]**
15. Is the line through $(1, 2)$ and $(4, 8)$ parallel to $y = 2x + 9$? Justify. **[L3]**

Perpendicular Lines (negative reciprocal slopes)

Example:

A line has slope 2. Find the slope of a line perpendicular to it.
Perpendicular slope is the negative reciprocal of 2.
 $= -1/2$. (Check: $2 \times (-1/2) = -1$.)

16. State the slope of a line perpendicular to a line with slope 3. **[L1]**

17. State the slope of a line perpendicular to $y = -4x + 1$. **[L1]**
18. A line has slope $2/3$. Find the slope of a line perpendicular to it. **[L2]**
19. Are $y = 2x + 1$ and $y = -1/2 x + 4$ perpendicular? Justify. **[L3]**

Length of a Line Segment (distance formula)

Example:

Find the length of the segment from A(1, 2) to B(4, 6).
 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9 + 16}$.
 $= \sqrt{25} = 5$. Answer: length = 5.

20. Find the length of the segment from (0, 0) to (6, 8). **[L1]**

21. Find the length of the segment from (1, 1) to (4, 5). **[L1]**
22. Find the length of the segment from (-2, 3) to (3, 15). **[L2]**
23. A triangle has vertices A(0, 0), B(6, 0) and C(6, 8). Find its perimeter. **[L3]**

Midpoint of a Segment

Example:

Find the midpoint of A(2, 4) and B(8, 10).

$$M = ((x_1 + x_2)/2, (y_1 + y_2)/2) = ((2+8)/2, (4+10)/2). \\ = (5, 7). \text{ Answer: midpoint } (5, 7).$$

24. Find the midpoint of (0, 0) and (6, 8). **[L1]**

25. Find the midpoint of $(1, 3)$ and $(7, 9)$. **[L1]**

26. Find the midpoint of $(-4, 2)$ and $(6, -8)$. **[L2]**

27. $M(5, 4)$ is the midpoint of $A(2, 1)$ and B . Find the coordinates of B . **[L3]**

Answers

Warm-up: W1) 4 W2) (0, 9) W3) 2

4. (L1) The coefficient of x is the slope: $m = 5$.
5. (L1) $m = (10 - 2)/(5 - 1) = 8/4 = 2$.
6. (L2) $m = (-8 - 4)/(3 - (-3)) = -12/6 = -2$.
7. (L3) $8/(6 - k) = 4$, so $6 - k = 2$, giving $k = 4$.
8. (L1) Slope = 1; y-intercept (0, 3).
9. (L1) y-intercept (0, 6); x-intercept: $0 = -2x + 6 \rightarrow x = 3 \rightarrow (3, 0)$.
10. (L2) Slope = 3; y-intercept (0, -6); e.g. (0, -6) and (2, 0).
11. (L3) Rearrange to $y = -2x + 4$: slope = -2; y-intercept (0, 4); points (0, 4) and (2, 0).
12. (L1) Parallel lines share the slope: $m = 4$.
13. (L1) Yes; both have slope 2, so they are parallel.
14. (L2) Same slope -3, y-intercept 2: $y = -3x + 2$.
15. (L3) Slope = $(8 - 2)/(4 - 1) = 6/3 = 2$, equal to 2, so yes — parallel.
16. (L1) Negative reciprocal: $-1/3$.
17. (L1) Given slope -4; perpendicular slope = $1/4$.
18. (L2) Negative reciprocal of $2/3$ is $-3/2$.
19. (L3) Product of slopes = $2 \times (-1/2) = -1$, so yes — the lines are perpendicular.
20. (L1) $d = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.
21. (L1) $d = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.
22. (L2) $d = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.
23. (L3) $AB = 6$, $BC = 8$, $CA = \sqrt{6^2 + 8^2} = 10$; perimeter = $6 + 8 + 10 = 24$.
24. (L1) $M = (3, 4)$.
25. (L1) $M = ((1+7)/2, (3+9)/2) = (4, 6)$.
26. (L2) $M = ((-4+6)/2, (2-8)/2) = (1, -3)$.
27. (L3) $B = (2 \cdot 5 - 2, 2 \cdot 4 - 1) = (8, 7)$.